



BUDGE BUDGE INSTITUTE OF TECHNOLOGY

Nischintapur, Budge Budge, Kolkata-700137

Department of Computer Science and Engineering(Cyber Security)

Academic Session: 2025-2026 (EVEN Semester)

Subject: DataBase Management Lab

Subject Code: PCC-CS692

SET-1

(A) Create the following tables with the mapping given below.

stu_details (reg_no, stu_name, DOB, address, city) mark_details (reg_no, mark1, mark2, mark3, total

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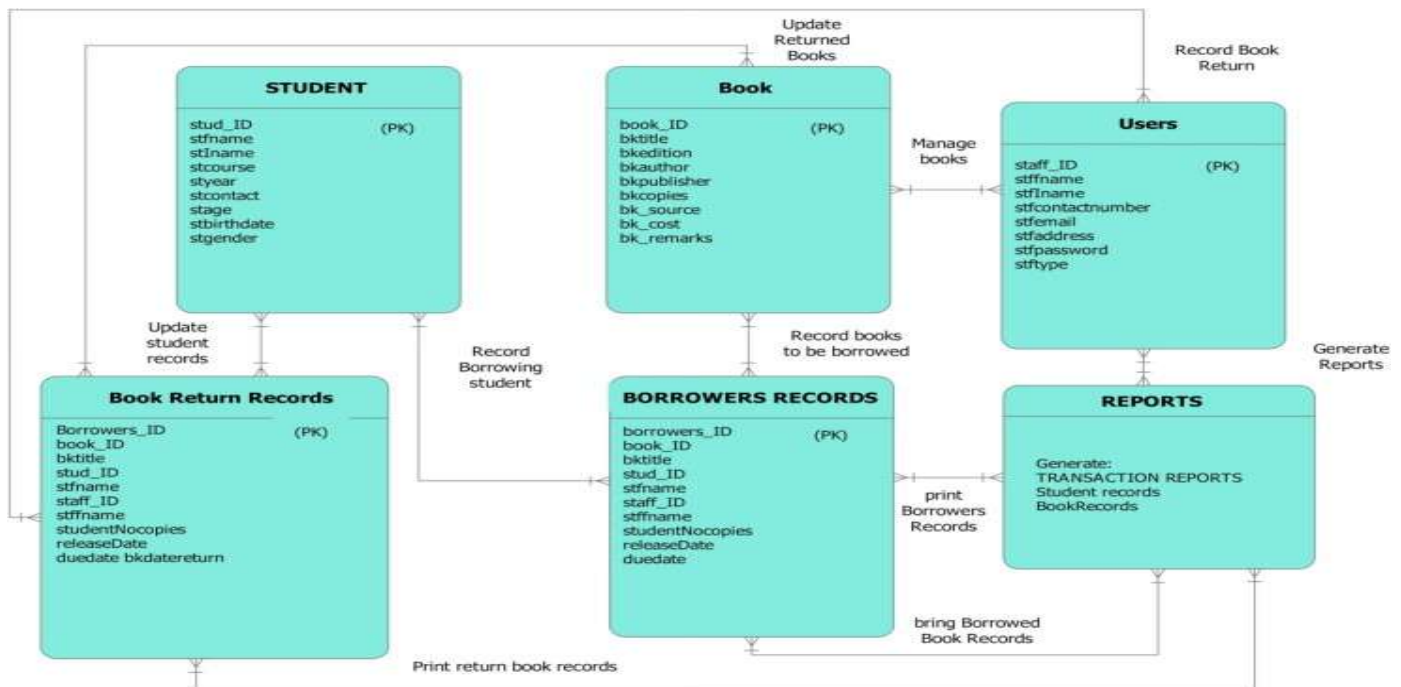
(i) Alter the table mark_details to add a column average with data type as long.

(ii) Display the months between the DOB and till date

(iii) Using alter command drop the column address from the table stu_details.

(B) Write a SQL program to find the sum & average marks of all the student using procedures.

(C)



Problem A college library uses the schema shown to manage students, staff users, books, borrowing, returns, and reports. Assume these tables and primary keys:

- STUDENT(stud_ID PK, stfname, stlname, stcourse, styear, stcontact, stage, stbirthdate, stgender)
- Users(staff_ID PK, stffname, stflname, stfcontactnumber, stfemail, stfaddress, stfpassword, stftype)
- Book(book_ID PK, bktitle, bkedition, bkauthor, bkpublisher, bkcopies, bk_source, bk_cost, bk_remarks)
- BORROWERS_RECORDS(borrowers_ID PK, book_ID FK, bktitle, stud_ID FK, stfname, staff_ID FK, stffname, studentNocopies, releaseDate, duedate)
- Book Return Records(Borrowers_ID PK/FK to BORROWERS_RECORDS, book_ID, bktitle, stud_ID, stfname, staff_ID, stffname, studentNocopies, releaseDate, duedate, bkdatereturn)
- REPORTS is a logical output, not a base table.

Business rules:

- A student can borrow at most 2 copies of the same book at any time.
- A book’s available copies = bkcopies – sum of currently borrowed (not yet returned) copies.
- A return is on time if bkdatereturn <= duedate; otherwise it is late and incurs a fee of 10 per calendar day late.

Tasks

1. When stud_ID = 20240123 borrows 1 copy of book_ID = 3105 on 2026-03-01, with a 14-day loan handled by staff_ID = 77, insert the borrowing record. Assume bktitle and names are redundantly stored as in the diagram.
2. Compute the available copies of book_ID = 3105 immediately after this borrowing, given Book.bkcopies = 5 and that, before this transaction, there was one active borrow of 1 copy for the same book.
3. On 2026-03-20, the same student returns the book to staff_ID = 80. Record the return.
4. Determine whether the return is late and calculate the late fee.
5. Produce a report query listing, for each student with any late returns in March 2026, the student ID, name, book title, days late, and fee, ordered by most days late.

Question Number.	Course Outcome(CO)	Taxonomy levels
Question No-1	CO1	3
Question No - 2	CO1	3
Question No - 3	CO3	6



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SET2

(A) Consider the following relations for a transport management system application:

BUS (ROUTENO, SOURCE, DESTINATION) DRIVER (DID, DNAME, DOB,

GENDER)ASSIGN_ROUTE (DID, ROUTENO, JOURNEY_DATE)

(i) Create the above mentioned tables and populate the tables.

(ii) Include constraints that the Route No starts with letter 'R' and gender of driver is always 'Male'.

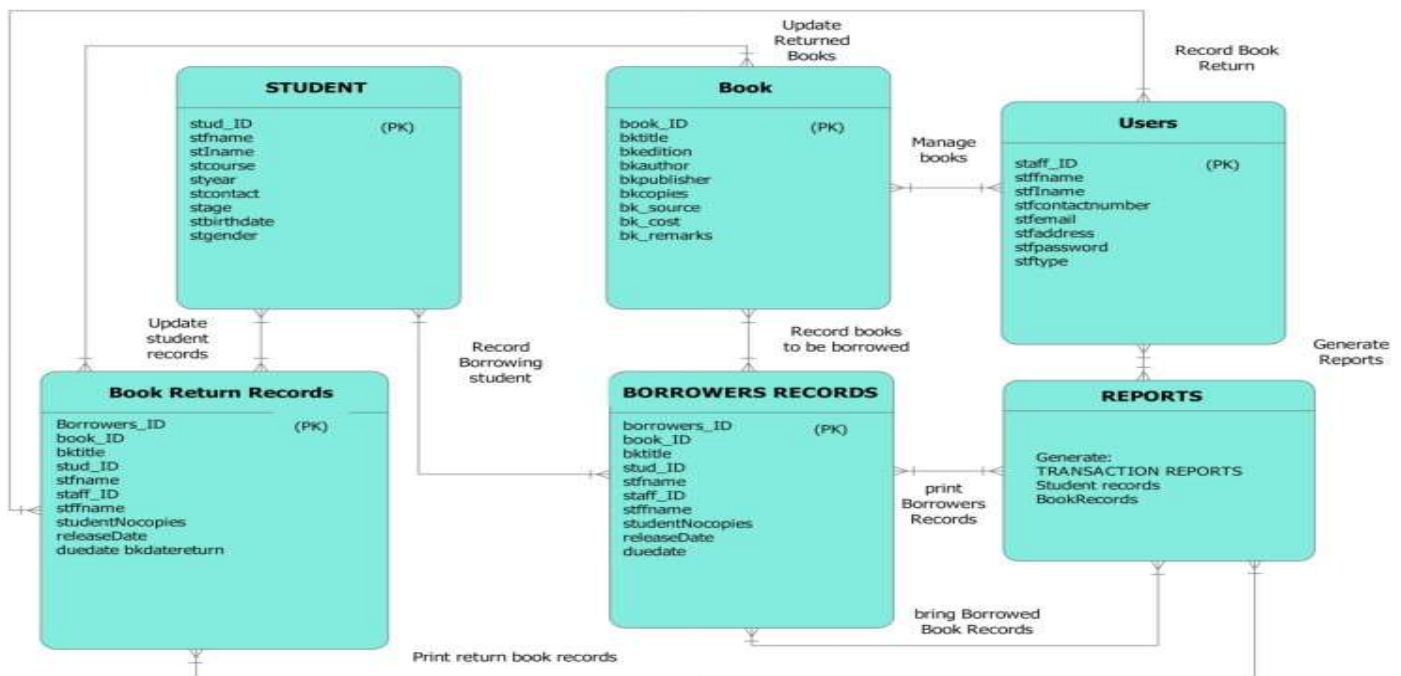
(iii) Develop a SQL query to list the details of drivers who have traveled more than three times on the same route

(iv) Create a sequence named Driver_ Sequence that will get incremented by 1. Use the created sequence while inserting DID into Driver table.

(v) Create a view that displays the DID, DNAME assigned for RouteNo 'R5'.

(B) Write a PL/SQL program to create a customer details using Cursor

(C)



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- Book(book_ID PK, bktitle, bkedition, bkauthor, bkpublisher, bkcopies, bk_source, bk_cost, bk_remarks)
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Question No-1	CO1	3
Question No - 2	CO1	4
Question No - 3	CO3	6



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SET 3

(A) Create the following tables with the mapping given below.

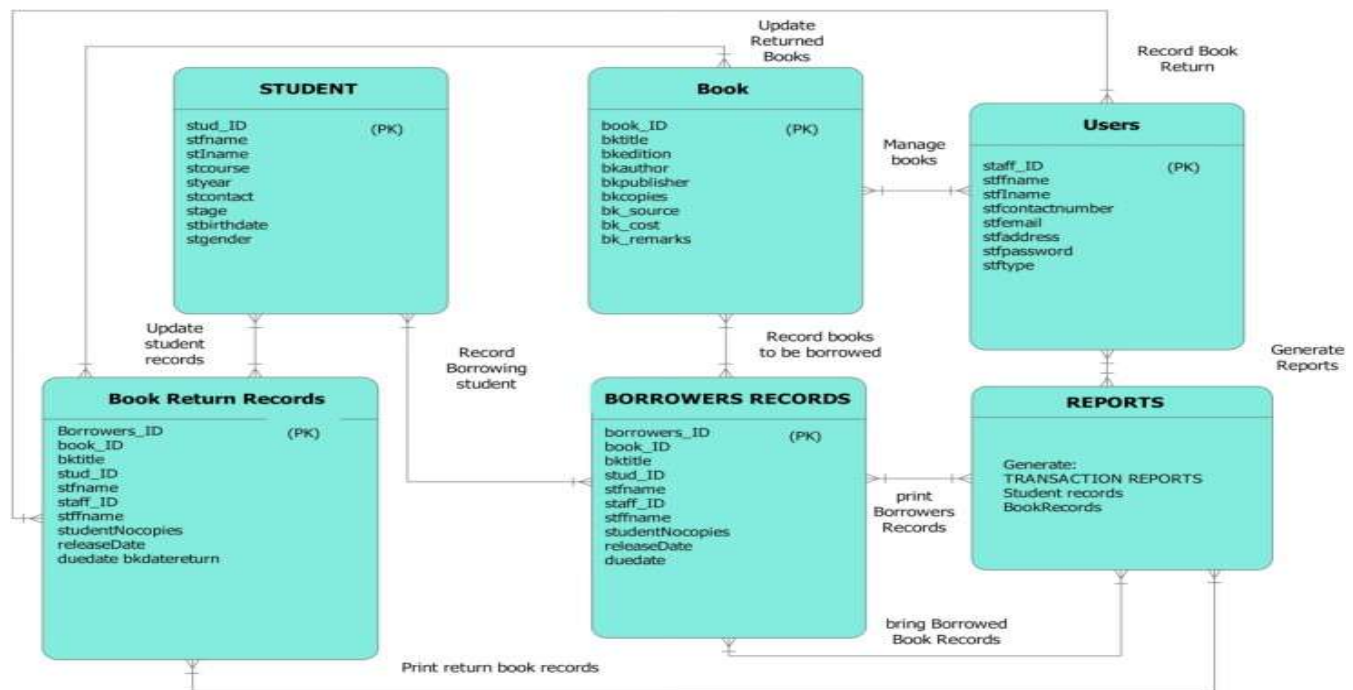
emp_details (emp_no, emp_name, DOB, address, mobile_no, dept_no, salary) dept_details (dept_no, dept_name, location)

(i) Select dept_no from dept_details and not in emp_details using both the tables.

(ii) Create a table named as student and insert values into the table.

(B) Write a trigger to avoid the entry of age less than 25.

(C)



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Tasks

1. When $\text{stud_ID} = 20240123$ borrows 1 copy of $\text{book_ID} = 3105$ on 2026-03-01, with a 14-day loan handled by $\text{staff_ID} = 77$, insert the borrowing record. Assume bktitle and names are redundantly stored as in the diagram.
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